

ForgeMind™

Showcase · S-MARATHON-22

VERSION 1.2.6 · OCTOBER 2026

Forged For Speed. Refined by Craft.

SEAGLASS FOUNDRY LLC

One scripted session, a complete search-and-rescue plan: 102 operations against real terrain, all succeeding — and exactly what the run does and does not show.

What You Are Looking At

S-MARATHON-22 is one continuous session of 102 requests planning the recovery of a downed pilot in the Galiuro Mountains of south-east Arizona, against real SRTM 1-arc-second elevation data. A rescue task force fixes the survivor and reads his terrain, defines the joint operations and search areas, plots the threats and a MANPADS site with their engagement zones, selects a helicopter landing zone that passes its slope gate and sits clear of the threat ring, plans a terrain-masked ingress, moves the ground team under overwatch, plans the egress and the casualty-evacuation handoff onto flat ground, and consolidates the recovery overlay on the military grid. It is a constructed training scenario, not a record of a real operation.

102	29	0	SRTM
requests in one continuous session, run top to bottom without a reset	named entities on the map at close-out, each created with explicit coordinates or derived from others	failed operations — every request routed, type-checked, executed and answered	real 1-arc-second elevation, probed in advance so the tactical decisions are real, not staged

How the run was made. This is a regression run from ForgeMind's test harness, not a live conversation. The harness supplies each turn's intent from the scenario script instead of asking a language model, and the map client is a test stand-in that records what would be drawn. What the session exercises is everything *after* the model: routing each request to a capability, type-checking it, resolving named entities across a hundred turns, and executing real geospatial analysis on real terrain. It says nothing about a model's understanding, or about model cost or latency. It was first run in June 2026, and the figures here are from its re-run in the regression sweep for 1.2.5, September 2026.

On 1.2.6. The session still runs 102 of 102 in the 1.2.6 regression corpus, with the same tactical outcome. ForgeGIS 1.2.6 measures geographic slope in ground metres on each axis, so slope readings move by under a degree (the survivor's ground reads 15.14° rather than 14.31°; *hlz_alpha* 16.66°, *hlz_bravo* 3.23°, the handoff 2.25° becomes 2.15°), and the masked routes re-trace slightly shorter (10.78 km ingress, 12.02 km egress). Every landing-zone and routing decision below is unchanged.

The Tactical Story

The layout sits on a west-to-east gradient — a steep ridge of 15–39° in the west, a flat bench of 1–5° in the east — so the decisions below are forced by the terrain, not written into the script.

Landing-zone selection is a real decision. Three candidates are tested against the stated gate — under 7° for a Black Hawk — and against the MANPADS ring. *hlz_alpha* at 15.96° fails on slope; *hlz_bravo* at 3.03° passes on slope but sits

inside the 4 km engagement zone; *hlz_charlie* at 1.12° and outside the ring is selected. The survivor himself is 3,287 m from the MANPADS, inside its ring, so the recovery is about moving him to ground that is clear.

Least-cost routing earns its place. The straight ingress is 7.17 km but crosses the engagement zone; the least-cost, terrain-masked ingress is 10.99 km — 3.8 km longer to fly the terrain. On egress the direct line is 9.60 km and clear; the masked route is 12.28 km.

The overwatch is sited honestly. From its knoll at 1,577 m it sees the survivor and is masked from the MANPADS, so it is not skylined. It cannot see the landing zone, which is defiladed behind the bench — exactly the kind of real visibility limit line-of-sight analysis exists to surface. The MANPADS likewise cannot see the survivor or the chosen landing zone, which is why the plan works.

The handoff lands on flat ground. The casualty-evacuation point sits at 2.25° on the valley floor.

Consolidation closes it out: the recovery hull, the combined threat picture — two disjoint 2 km rings and the 4 km MANPADS ring, 75 km² in all — the bounding box, and UTM zone 12N coordinates for the nine-line.

What It Exercises

- **Sustained coherence.** A hundred and two dependent requests in one session, with no context reset and no entity drift: 29 named entities built up and referenced by name, turn after turn.
- **Geospatial depth.** Elevation, slope, hillshade, contours, viewshed and line of sight, least-cost paths, buffers and corridors, unions, convex hull, centroid, containment and overlap tests, and reprojection to UTM — run on real terrain. A quarter of the requests loaded a raster.
- **The compiled path.** Creates of named points, polygons and lines run on the scene-entity lane; derived entities — threat rings, corridors, least-cost routes, the hull and the unions — and every value and analysis request run through the compiler to a validated pipeline.

The Full Session

Results are as the run logged them — run on 1.2.5, September 2026.

#	Request	Result
1	Mark the survivor's last known position at -110.245, 32.560 and call it survivor	created **survivor**
2	What's the elevation at survivor?	**1,602 m**
3	How steep is the ground at survivor? I need to know if he's on a slope.	**14.31°** (mid-slope)
4	Generate a hillshade around survivor so we can read the terrain he's in.	hillshade raster (361×361)
5	Generate a slope raster around survivor.	slope raster
6	Generate contours around survivor every 20 meters for the recovery overlay.	**535 contour lines** (1400–2260 m)
7	What's the highest ground within 2 kilometers of survivor?	max **1,912 m** (min 1,479 / mean 1,595; 20305 px)
8	Find the lowest ground within 1 kilometer of survivor.	min **1,505 m** (max 1,717 / mean 1,573; 5005 px)
9	Create a polygon … and call it joa	created **joa**
10	Create a polygon … and call it search_area	created **search_area**
11	What's the area of the joa in square kilometers?	**189.6 km²**
12	What's the area of search_area in square kilometers?	**20.84 km²**
13	Is survivor inside search_area?	**1/1 inside**
14	Is survivor inside the joa?	**1/1 inside**
15	Plot a suspected enemy position at -110.215, 32.540 and call it threat_1	created **threat_1**
16	Plot another suspected enemy position at -110.205, 32.590 and call it threat_2	created **threat_2**
17	Mark a possible MANPADS site on the ridge at -110.278, 32.550 and call it manpads	created **manpads**
18	What's the elevation at manpads? If they're up high it changes our ingress.	**2,083 m** (commanding ridge)
19	How far is threat_1 from survivor, in meters?	**3,586 m**
20	How far is threat_2 from survivor, in meters?	**5,018 m**
21	How far is the manpads site from survivor, in meters?	**3,287 m** (inside the 4 km WEZ)
22	Compute the viewshed from manpads at a 2 meter height. …	viewshed raster
23	Compute the viewshed from threat_1 at a 2 meter height.	viewshed raster
24	Is survivor visible from manpads? Can they see our guy?	**No** (0/1) — survivor is masked from the MANPADS
25	Does the manpads site have line of sight to threat_1? …	**No** (0/1) — not mutually supporting

#	Request	Result
26	Put a 4 kilometer ring around manpads and call it manpads_wez	created **manpads_wez** — buffer 4000 m (UTM 32612)
27	What's the area of manpads_wez in square kilometers?	**50.01 km²** (= π·4²)
28	Put a 2 kilometer ring around threat_1 and call it threat_1_ring	created **threat_1_ring** — buffer 2000 m
29	Put a 2 kilometer ring around threat_2 and call it threat_2_ring	created **threat_2_ring** — buffer 2000 m
30	Does survivor fall inside manpads_wez?	**1/1 inside** — our guy is under the threat
31	Does search_area overlap manpads_wez?	**Yes** (overlap)
32	We need an HLZ … out of the threat ring. Mark a candidate at -110.260, 32.560 and call it hlz_alpha	created **hlz_alpha**
33	Mark a second HLZ candidate at -110.245, 32.545 and call it hlz_bravo	created **hlz_bravo**
34	Mark a third candidate at -110.230, 32.560 and call it hlz_charlie	created **hlz_charlie**
35	What's the slope at hlz_alpha? It needs to be under 7 degrees for the Black Hawk.	**15.96° → reject** (fails the <7° gate)
36	What's the slope at hlz_bravo?	**3.03°** (passes the slope gate)
37	What's the slope at hlz_charlie?	**1.12°** (passes the slope gate)
38	What's the elevation at hlz_alpha?	**1,629 m**
39	What's the elevation at hlz_bravo?	**1,549 m**
40	What's the elevation at hlz_charlie?	**1,498 m**
41	Is hlz_alpha inside manpads_wez? If it's in the threat ring it's out.	**1/1 inside** (also exposed)
42	Is hlz_bravo inside manpads_wez?	**1/1 inside → reject** (in the threat ring)
43	Is hlz_charlie inside manpads_wez?	**0/1 — outside → SELECT** (flat + clear)
44	Can the manpads site see hlz_alpha?	**No** (0/1)
45	Can the manpads site see hlz_charlie?	**No** (0/1) — the chosen LZ is not observed by the MANPADS
46	How far is hlz_alpha from survivor, in meters? …	**1,406 m**
47	How far is hlz_charlie from survivor, in meters?	**1,406 m**
48	Range and bearing from hlz_charlie to survivor.	range **1,409 m**, bearing **270.0°** (due west)
49	Set the ingress start point at -110.290, 32.520 and call it ingress_ip	created **ingress_ip**
50	What's the elevation at ingress_ip?	**1,877 m**
51	Create a line from -110.290, 32.520 to -110.230, 32.560 and call it ingress_route	created **ingress_route** (straight line)

#	Request	Result
52	How long is ingress_route in kilometers?	**7.171 km** (direct)
53	Pull an elevation profile along ingress_route. Let me see what the direct run looks like.	128-station profile, **1,498–2,056 m**
54	Does ingress_route pass through manpads_wez? If it does we reroute.	**Yes** (crosses) — direct route is exposed → reroute
55	Does ingress_route cross threat_1_ring?	**No** (clear of threat_1)
56	Compute the least cost path from ingress_ip to hlz_charlie and call it masked_ingress	created **masked_ingress** — `LEAST_COST_PATH` (354 cells)
57	How long is masked_ingress in kilometers?	**10.99 km** (masked; +3.8 km to fly the terrain)
58	Pull an elevation profile along masked_ingress. I want to fly it nap-of-the-earth.	128-station profile, **1,490–2,080 m**
59	Buffer masked_ingress by 500 meters … and call it ingress_corridor	created **ingress_corridor** — line buffer 500 m
60	Does ingress_corridor overlap manpads_wez?	**Yes** (overlap)
61	The ground team inserts at the HLZ. Create a line … and call it gt_route	created **gt_route** (straight line)
62	How long is gt_route in meters?	**1,406 m**
63	Pull an elevation profile along gt_route. How much climbing for the team carrying gear?	128-station profile, **1,498–1,619 m** (~120 m of relief)
64	Does gt_route cross into manpads_wez?	**Yes** (crosses) — the team enters the ring to reach the survivor
65	Buffer gt_route by 100 meters … and call it gt_corridor	created **gt_corridor** — line buffer 100 m
66	Mark an overwatch position for the team at -110.245, 32.555 and call it gt_overwatch	created **gt_overwatch**
67	What's the elevation at gt_overwatch?	**1,577 m**
68	Compute the viewshed from gt_overwatch at a 2 meter height. I want eyes on the survivor and his draw.	viewshed raster
69	Can gt_overwatch see survivor?	**Yes** — eyes on the survivor
70	Can gt_overwatch see hlz_charlie?	**No** — LZ defiladed behind the bench shelf (inbound aircraft owns the LZ)
71	Does gt_overwatch have line of sight to manpads? I don't want them skylined to that site.	**No** — masked from the MANPADS (not skylined)
72	Set the egress endpoint at -110.170, 32.630 and call it egress_rp	created **egress_rp**
73	What's the elevation at egress_rp?	**1,331 m** (valley floor)

#	Request	Result
74	Create a line from -110.230, 32.560 to -110.170, 32.630 and call it egress_route	created **egress_route** (straight line)
75	How long is egress_route in kilometers?	**9.601 km** (direct)
76	Does egress_route pass through manpads_wez? We are not flying back out through the threat.	**No** — clear of the WEZ
77	Compute the least cost path from hlz_charlie to egress_rp and call it masked_egress	created **masked_egress** — `LEAST_COST_PATH` (404 cells)
78	How long is masked_egress in kilometers?	**12.28 km** (masked)
79	Pull an elevation profile along masked_egress.	128-station profile, **1,303–1,501 m**
80	Does masked_egress cross threat_2_ring?	**Yes** (crosses) — flag the threat on the egress
81	Buffer masked_egress by 500 meters … and call it egress_corridor	created **egress_corridor** — line buffer 500 m
82	Mark the CASEVAC handoff point at -110.165, 32.625 and call it casevac_hp	created **casevac_hp**
83	What's the distance from egress_rp to casevac_hp, in kilometers?	**727 m** (0.727 km)
84	What's the slope at casevac_hp? Confirming the second bird can set down.	**2.25°** — flat, the second bird can set down
85	Create a line from -110.170, 32.630 to -110.165, 32.625 and call it casevac_leg	created **casevac_leg** (straight line)
86	How long is casevac_leg in kilometers?	**0.727 km**
87	Find the centroid of search_area and call it search_center	created **search_center** (centroid)
88	How far is search_center from survivor, in meters?	**≈ 0 m** — search_center coincides with survivor by construction
89	Draw the convex hull around survivor, hlz_charlie, gt_overwatch, and ingress_ip and call it recovery_hull	created **recovery_hull** (convex hull, 4 pts)
90	What's the area of recovery_hull in square kilometers?	**3.127 km²**
91	Union threat_1_ring and threat_2_ring into combined_threat	created **combined_threat** — `DISSOLVE_ENTITIES` (2 feat — disjoint)
92	What's the area of combined_threat in square kilometers?	**25.00 km²** (≈ 2 × π·2²)
93	Union manpads_wez and combined_threat into total_threat	created **total_threat** — `DISSOLVE_ENTITIES` (3 feat — disjoint)
94	What's the area of total_threat in square kilometers?	**75.01 km²** (= 50.0 + 12.5 + 12.5, clean)
95	Does recovery_hull overlap total_threat? …	**Yes** (overlap)
96	What's the bounding box of survivor, hlz_charlie, gt_overwatch, and egress_rp?	W -110.245 / S 32.555 / E -110.17 / N 32.63 (4 pts)

#	Request	Result
97	Reproject survivor to UTM zone 12N for the nine-line.	**EPSG:32612** (570878, 3602762)
98	Reproject hlz_charlie to UTM zone 12N.	**EPSG:32612** (572286, 3602772)
99	Reproject masked_ingress to UTM zone 12N.	**EPSG:32612** (REPROJECT_LAYER, 1 feat)
100	Reproject recovery_hull to UTM zone 12N.	**EPSG:32612** (REPROJECT_LAYER, 1 feat)
101	Generate contours around hlz_charlie every 10 meters for the HLZ card.	**1143 contour lines** (1360–2250 m)
102	List everything on the map so I can confirm the recovery package is complete.	lists all **29 entities** on the map

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